

Safa MESSAOUD

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LinkedIn: <https://www.linkedin.com/in/safa-messaoud-ph-d-638444144/>

Google Scholar: <https://scholar.google.com/citations?user=M2wXrP4AAAAJ&hl=en>

RESEARCH INTERESTS

Deep Learning, Generative models, Large Language Models, Reinforcement Learning, Computer Vision, Healthcare Analytics, Safe AI

WORK EXPERIENCE

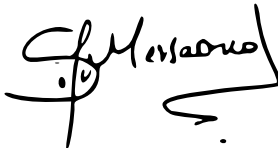
- AI Scientist**2022-Present
- Qatar Computing Research Institute, Qatar*
- Co-Founder & Co-Director, [MenaML](#)**2025-Present
- Co-founded and leads a regional machine learning education and mentorship initiative across the MENA region*

EDUCATION

- Doctor of Philosophy in Electrical and Computer Engineering**2014-2021
- University of Illinois at Urbana-Champaign, USA*
- Advisor: Prof. Alexander G. Schwing
- Thesis: Toward More Scalable Structured Models
- GPA: 3.8/4.0
- Master of Science in Computer Engineering**2013-2014
- Virginia Polytechnic and State University, USA*
- Advisor: Prof. Sandeep Shukla
- MS Thesis: Translating discrete time SIMULINK to SIGNAL
- GPA: 4.0/4.0
- Visiting Researcher, Concurrent Enrollment Student**2012-2013
- University of California Berkeley, USA*
- Advisors: Prof. Alberto Sangiovanni Vincentelli (UC Berkeley), Prof. Andreas Herkersdorf (TUM)
- MS Thesis: Optimal Architecture Synthesis of Aircraft Electrical Power Systems
- GPA: 3.85/4.0
- Master of Science in Electrical Engineering and Information Technology**2011-2013
- Technical University of Munich, GERMANY*
- MS Thesis conducted at the University of California Berkeley
- GPA: 4.0/4.0 (German 1.5)
- Bachelor of Science in Electrical Engineering and Information Technology**2008-2011
- Technical University of Munich, GERMANY*
- Advisor: Dr. Martin Brischwein
- BS Thesis: Validation of a Finite Element Model for Reaction and Diffusion Using Electro-Chemical and Signal-Processing Methods
- GPA: 4.0/4.0 (German 1.3)
- Pre-Engineering School Certificate**2007-2008
- Leibniz University of Hannover, GERMANY*
- GPA: 4.0/4.0 (German 1.0)
- Baccalauréat en Mathématiques (High school degree)**2003-2007
- Elite High School “Lycée Pilote de Sfax”, TUNISIA*
- Rank: 6th nationally
- GPA: 19.54/20

PUBLICATIONS

- Can We Compute The Entropy Of Arbitrary Distributions Known Up To A Normalization Constant? A Tale of SVGD Scalability**
- [Safa Messaoud](#), Skander Charni, Elaa Bouazza, Ali Pourghasemi, Halima Bensmail*
- Under submission
- Explaining the role of Intrinsic Dimensionality in Adversarial Training**
- Enes Altinisik, [Safa Messaoud](#), Taha Sencar, Hassan Sajjad, Sanjay Chawla*
- ICML 25
- S2AC: Energy-Based Reinforcement Learning with Stein Soft Actor Critic**
- [Safa Messaoud](#), Billel Mokeddem*, Zhenghai Xue*, Linsey Pang, Bo An, Haipeng Chen, Sanjay Chawla*
- ICLR 24
- A3T: Accuracy-Aware Adversarial Training**
- Enes Altinisik, [Safa Messaoud](#), Husrev Taha Sencar, Sanjay Chawla*
- Mach. Learn. 23 (among top 3 papers)
- Impact of adversarial training on robustness and generalizability of language models**
- Enes Altinisik, Hassan Sajjad, Husrev Taha Sencar, [Safa Messaoud](#), Sanjay Chawla*
- Findings of ACL 22
- DeepQAMVS: Query-Aware Hierarchical Pointer Networks for Multi-Video Summarization**
- [Safa Messaoud](#), Issmini Lourentzou, Assma Boughoula, Mona Zehni, ChangXiang Zhai, Zhizhen Zhao, Alexander G. Schwing*
- ACM SIGIR 21
- Can We Learn Heuristics for Graphical Model Inference Using Reinforcement Learning?**
- [Safa Messaoud](#), Maghav Kumar, Alexander G. Schwing*
- CVPR 20 (Oral)
- Structural Consistency and Controllability for Diverse Colorization**
- [Safa Messaoud](#), David A. Forsyth, Alexander G. Schwing*
- ECCV 18



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Google Scholar: <https://scholar.google.com/citations?user=M2wXrP4AAAAJ&hl=en>

Unsupervised Analysis of Transcriptomic Data for Demystifying the Cause of Alzheimer Disease **AAIC 17**
Yogatheesan Varatharajah, Mariet Younkin, Xue Wang, Arjun Athreya, Safa Messaoud, Ravi Iyer, Nilufer Ertekin-Taner

Translating Discrete Time SIMULINK to SIGNAL **e-TI 15**
Safa Messaoud, Neda Saeedloei, Sandeep Shukla

Reaction–Diffusion Modelling for Microphysiometry on Cellular Specimens **Med. Biol. Eng. Comput 13**
Daniel Grundl, Xiaorui Zhang, Safa Messaoud, Cornelia Pfister, Franz Demmel, Mario S. Mommer, Bernhard Wolf, and Martin Brischwein

PATENTS

Medical Record Problem List Generation **U.S. Patent**
Murthy V. Devarakonda, Safa Messaoud, Ching-Huei Tsou

Accelerating Genomic Data Parsing on Field Programmable Gate Arrays **U.S. Patent**
Safa Messaoud, Takeshi Ogasawara

TECHNICAL REPORTS

Fanar: An Arabic-Centric Multimodal Generative AI Platform." arXiv preprint arXiv:2501.13944 (2025). Team, Fanar, et al.

Toward More Scalable Structured Models **PhD Thesis**

Translating Discrete-Time SIMULINK to SIGNAL **MS Thesis**

Optimal Architecture Synthesis for Aircraft Electrical Power Systems **MS Thesis**

Validation Study of a Finite Element Model for Reaction and Diffusion Using Electro-Chemical and Mathematical Methods **BS Thesis**

GRANTS

Social Innovation Fund 8th cycle (Amount: QAR 50,000)
Project: Mentorship Platform for professional roadmaps building, sharing and merging
Role: Lead Principal Investigator

Signature Research Grant (HBKU) for advancing Qatar's AI landscape via developing an Arabic-centric and ethically aligned Large Language Model (Amount: QAR 600,000)
Role: Principal Investigator

TALKS AND PANELS

Diffusion Models (1h) **2025**
Venue: Middle East and North Africa Machine Learning (MenaML) Winter School

Panelist (1h) **2024**
Venue: Executive Panel by PwC and Microsoft on “Women leading transformation Qatar”, Qatar

Generative AI (2h) **2023**
Venue: AI Summer School at Texas A&M University, Qatar

Exploring ChatGPT and Beyond for Enhanced Workspace Productivity (45 min) **2023**
Venue: Dahua Technology Workshop on AI and IoT technology

Score Matching (30 min) **2021**
Venue: Vision Lunch, UIUC

Meta-Algorithms (30 min) **2017**
Venue: CS 598 LAZ: Cutting-Edge Trends in Deep Learning and Recognition course, UIUC

Polychronous Formal Languages for Model-Based Design (30 min) **2015**
Venue: Arab Woman in Computing conference (Invited)

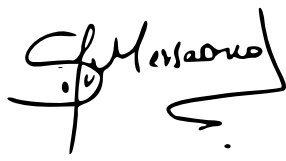
Methods for Designing Optimized Architectures for Aircraft Electrical Power Systems (45 min) **2014**
Venue: Weekly research seminar of the ECE department at Virginia Tech (Invited)

TEACHING

Pattern Recognition ECE 544, University of Illinois at Urbana-Champaign
Teaching Assistant (Fall 2017, Fall 2018); *Head Teaching Assistant* (Fall 2019, Fall 2020)

Machine Learning CS 446, University of Illinois at Urbana-Champaign
Teaching Assistant (Spring 2018, Spring 2019, Spring 2021); *Head Teaching Assistant* (Spring 2020)
Tasks: Gave lecture on Reinforcement Learning (Fall 2018, Fall 2019), held office hours (1h a week), answered Piazza questions, participated in exams and homeworks design, graded homeworks, exams and projects. As head teaching assistant, coordinated between other teaching assistants and graders for exams design and grading tasks.

Digital Design Lab ECE 3544, Virginia Polytechnic and State University
Teaching Assistant (Fall 2013)
Tasks: Held lab office hours (9h a week) and graded homeworks.



LEADERSHIP AND SERVICE

(Meta) reviewer, ECCV, NeurIPS, AAAI, ICLR	2023, 2024, 2025
Machine Learning and Data Science Symposium, QCRI Co-chair	2024
The National Scientific Research and Innovation Week, Qatar Judge	2023
CSL Student Conference, University of Illinois at Urbana-Champaign Chair of the <i>Artificial Intelligence in Action</i> session	2018
CSL Student Conference, University of Illinois at Urbana-Champaign Chair of the <i>Computational Genomics</i> session	2017
Children Parliament of Tunisia Head of the Children Protection Committee	2003-2005

HONORS

- HBKU Reyada reward 2024 (Entrepreneurship program)
- Travel grant from the ECE department at UIUC to attend NeurIPS 2017.
- CompGen fellowship at UIUC (2015-2016).
- Arab Woman in Computing scholarship to attend ArabWIC 2015 conference.
- The Grace Hooper scholarship for women in computing, 2015.
- Graduate Woman Cohort scholarship, 2014-2015.
- The Heinrich and Muhlfenzl scholarship for exchange students, 2012-2013.
- Automotive, Railway and Avionic Multicore Systems grant from the German research ministry (Funding my research stay in Berkeley in 2013).
- Scholarship awarded by the Tunisian government and the DAAD for Bachelor and Master degrees in Germany for ranking among the top 10 students in Tunisia in the national high school examination (2007-2013)
- German Pedagogic Exchange Service (PAD) international award for winning the competition of the best German-speaking pupils in Tunisia (selected among 3 pupils from Tunisia), 2006
- Prize of the regional "Journée du Savoir" in Sfax, Tunisia, awarded three times (7th, 8th, 10th grades) for ranking 1st in the governorate of Sfax.

RESEARCH

Generative Models + Reinforcement Learning

- Extended Gaussian conditional random fields, traditionally unimodal and only capturing pairwise variables interactions, to model multi-modal distributions with high-order dependencies, while enabling exact inference and incorporating external constraints at runtime, for the task of diverse image colorization. **ECCV 18, WiML 17**
- Developed a reinforcement learning based engine for solving inference in energy-based models with traditionally intractable higher-order potentials, for semantic segmentation. **CVPR 20 (oral), WiCV 20 (oral), DeepVision 20**
- Designed a reinforcement learning based method to train a pointer networks model with hierarchical attentions, achieving state-of-the-art results on query-aware multi-video summarization. **ACM SIGIR 21, DRL4IR 21 (oral)**
- Proposed a news variational distribution leveraging Stein Variational Gradient Descent dynamics that led to learning multi-modal policies in the context of Max-Entropy Reinforcement Learning. **ICLR 24**

Safe AI (Adversarial Training)

- Improved the generalization/robustness tradeoff in adversarial training via leveraging the intrinsic dimensionality of the data manifold and the misclassification accuracy. **Mach. Learn. 23 (selected among top 3 papers), Findings of ACL 22**

Healthcare Analytics

- Developed disease-category based models for problem list generation from electronic health records. Used several variations of autoencoders to discover customized features for disease categories. **U.S. Patent**
- Proposed an unsupervised technique to identify genes that discriminate temporal cortex expression data of Alzheimer affected patients from control subjects. **AAIC 17**
- Extended a 3D finite element simulation of diffusion and metabolic reaction of cellular specimens with a model of the sensor effect. Validated the framework via an electro-chemical experiment. **Med. Biol. Eng. Comput 13**

Automated Tools for Cyber-Physical Systems Design

- Developed two optimization-oriented methodologies to synthesize a cost-effective and reliable aircraft electrical power systems topology: (1) Mixed Integer-Linear Programming modulo reliability, and (2) Integer-Linear programming with approximate reliability algebra. **MS Thesis**
- Devised a semantic translator that transforms discrete time SIMULINK models into SIGNAL programs. SIGNAL is a formal language that enables correct by design and multi-thread code generation. **MS Thesis**

Computer Architecture

- Designed an FPGA based accelerator for genomic files parsing that resulted in 10x speed-up compared to the software implementation. **U.S. Patent**
- Designed an FPGA based monitor aggregator for multicore architecture resource-aware scheduling, within the scope of the HW architecture paradigm “Invasive Computing”

Shirvan

INTERNSHIPS

Research Intern, Medical Records Analysis Group, IBM Watson, USA **May-Aug 2016**

Manager: Dr. Murthy Devarakonda; *Mentor:* Dr. Ching-Huei Tsou

- Proposed a novel algorithm for disease-based problem list generation from electronic medical records (U.S. Patent)

Research Intern, High Performance Commercial Systems at IBM Research-Tokyo, JAPAN **May-Aug 2015**

Manager: Dr. Morioshi Ohara; *Mentor:* Dr. Takeshi Ogasawara

- Designed an FPGA-based accelerator for genomic files parsing (SAM to BAM conversion)
- Achieved 10x Speedup compared to the single threaded SW implementation (U.S. Patent)

Research Intern, Department of Integrated Systems at the Technical University of Munich, GERMANY

Professor: Dr. Andreas Herkersdorf; *Mentor:* Dr. Ravi Kumar Pujari

Aug-Sep 2012

- Designed an FPGA-based monitor aggregator for multicore architecture resource-aware scheduling, within the scope of the HW architecture paradigm “Invasive Computing”

Research Intern, SIEMENS Corporate Technology in Munich, GERMANY

March-May 2011

Mentor: Dr. Matthias Gerlich

- Implemented a data acquisition system for electric motor testing using MATLAB

Intern, Institute of Energy Conversion System at the TUM, GERMANY

May-Oct 2010

- Built a buggy-based electric car “eCARus” with other students
- Designed printed circuit boards of the battery management system

Intern, SIEMENS Corporate Technology in Munich, GERMANY

July-Oct 2008

- Pre-engineering internship: basics of IT

TECHNICAL SKILLS

Programming Languages (SW): Python, MATLAB, C, C++

Programming Languages (HW): Verilog, VHDL, OpenCL

Deep Learning Frameworks: Pytorch, TensorFlow

LANGUAGES

Arabic (fluent), French (fluent), English (fluent) and German (fluent)

HOBBIES

Reading, Painting, Playing Piano, Traveling

